

CHAPTER 1

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
```

```
faheemahazin5@cloudshell:~$ uname
```

```
Linux
```

```
faheemahazin5@cloudshell:~$ uptime
```

```
10:15:57 up 3:42, 0 user, load average: 0.50, 0.15, 0.05
```

```
faheemahazin5@cloudshell:~$ hostname
```

```
cs-782581850354-default
```

```
faheemahazin5@cloudshell:~$ dmesg
```

```
dmesg: read kernel buffer failed: Operation not permitted
```

```
faheemahazin5@cloudshell:~$ free
```

	total	used	free	shared	buff/cache	available
Mem:	16382836	851676	14445696	1136	1364324	15531160
Swap:	0	0	0			

```
faheemahazin5@cloudshell:~$
```

```
Tasks: 28 total, 1 running, 27 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15998.9 total, 14123.9 free, 814.1 used, 1333.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 15184.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
477	root	20	0	1867212	46076	32012	S	0.3	0.3	0:00.49	containerd
1503	faheema+	20	0	11.3g	96200	50576	S	0.3	0.6	0:01.12	node
1	root	20	0	4324	3260	2976	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3812	2464	S	0.0	0.0	0:00.01	rsyslogd
25	root	20	0	38768	28020	10920	S	0.0	0.2	0:00.49	python
26	root	20	0	6196	2620	2380	S	0.0	0.0	0:00.00	logger
67	root	10	-10	12024	4088	2972	S	0.0	0.0	0:00.00	sshd
325	root	20	0	2161192	84668	60884	S	0.0	0.5	0:00.35	dockerd
436	root	20	0	1234524	12732	8268	S	0.0	0.1	0:00.19	editor-proxy
437	root	20	0	16328	6416	5492	S	0.0	0.0	0:00.00	sudo
440	root	20	0	1226100	2744	1976	S	0.0	0.0	0:00.00	tmux-agent
444	root	10	-10	13996	9684	8316	S	0.0	0.1	0:00.02	sshd
482	root	10	-10	13996	9656	8288	S	0.0	0.1	0:00.02	sshd
628	faheema+	10	-10	14256	6352	4700	S	0.0	0.0	0:00.00	sshd
672	faheema+	10	-10	14256	6348	4696	S	0.0	0.0	0:00.00	sshd
690	root	20	0	2696	1652	1552	S	0.0	0.0	0:00.00	sleep
694	faheema+	10	-10	7340	3652	3408	S	0.0	0.0	0:00.00	bash
697	faheema+	10	-10	7604	4444	3836	S	0.0	0.0	0:00.00	bash
699	root	20	0	9196	3988	3612	S	0.0	0.0	0:00.00	runuser
702	faheema+	20	0	4324	3188	2924	S	0.0	0.0	0:00.00	sh
723	faheema+	20	0	11.3g	133104	50948	S	0.0	0.8	0:05.07	node
724	faheema+	10	-10	7340	3664	3416	S	0.0	0.0	0:00.00	bash
725	faheema+	10	-10	7340	3712	3452	S	0.0	0.0	0:00.00	start-shell.sh
726	faheema+	10	-10	6880	3216	2912	S	0.0	0.0	0:00.00	tmux
728	faheema+	10	-10	7148	2628	1996	S	0.0	0.0	0:00.01	tmux
729	faheema+	10	-10	8796	5916	4048	S	0.0	0.0	0:00.05	bash
1275	faheema+	20	0	1048516	61368	47528	S	0.0	0.4	0:00.25	node
1561	faheema+	10	-10	11928	5708	3492	R	0.0	0.0	0:00.00	top

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	4324	3260	2976	S	0.0	0.0	0:00.05	/bin/bash /google/scripts/onrun.sh sleep infinity
487	root	20	0	18236	45952	2012	S	0.7	0.3	0:00.03	containerd --config /var/run/docker/containers/containerd.toml
494	root	20	0	18236	45952	2012	S	0.7	0.3	0:00.07	containerd --config /var/run/docker/containers/containerd.toml
728	faheemahaz	10	-10	7280	2744	1996	S	0.7	0.0	0:00.03	tmux new-session -A -D -n cloudshell -s 637258788
1	root	20	0	4324	3260	2976	S	0.0	0.0	0:00.05	/bin/bash /google/scripts/onrun.sh sleep infinity
25	root	20	0	39768	20200	1920	S	0.0	0.2	0:00.52	/usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisord.pid --logfile=/var/
26	root	20	0	8196	2620	3380	S	0.0	0.0	0:00.00	logger -t supervisord
67	root	10	-10	12024	4088	2972	S	0.0	0.0	0:00.00	sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
325	root	20	0	21108	84796	61012	S	0.0	0.5	0:00.18	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
399	root	20	0	21108	84796	61012	S	0.0	0.5	0:00.03	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
401	root	20	0	21108	84796	61012	S	0.0	0.5	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
402	root	20	0	21108	84796	61012	S	0.0	0.5	0:00.01	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
404	root	20	0	21108	84796	61012	S	0.0	0.5	0:00.01	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
406	root	20	0	21108	84796	61012	S	0.0	0.5	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
435	root	20	0	21108	84796	61012	S	0.0	0.5	0:00.00	/usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
436	root	20	0	12036	12536	2688	S	0.0	0.1	0:00.06	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
437	root	20	0	18228	4416	4492	S	0.0	0.1	0:00.00	sudo /google/devshell/tmux-agent
439	root	20	0	12036	12536	2688	S	0.0	0.1	0:00.03	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
440	root	20	0	11976	2744	1976	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
441	root	20	0	12036	12536	2688	S	0.0	0.1	0:00.00	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
442	root	20	0	12036	12536	2688	S	0.0	0.1	0:00.00	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
443	root	20	0	12036	12536	2688	S	0.0	0.1	0:00.02	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
444	root	10	-10	13996	9684	8316	S	0.0	0.1	0:00.02	sshd: faheemahazin5 [priv]
445	root	20	0	11976	2744	1976	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
446	root	20	0	11976	2744	1976	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
447	root	20	0	11976	2744	1976	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
448	root	20	0	11976	2744	1976	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
451	root	20	0	12036	12536	2688	S	0.0	0.1	0:00.05	/google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
477	root	20	0	18236	45952	2012	S	0.0	0.3	0:00.05	containerd --config /var/run/docker/containers/containerd.toml

CHAPTER 2

```
Cloud Shell Editor

cloudshell X + ▾

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info$ mkdir chapter2_file_management
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info$ cd chapter2_file_management
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management$ mkdir demo_folder

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management$ mkdir project_data

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management$ mkdir backup_folder
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management$ pwd

/home/faheemahazin5/linux_lab/chapter1_system_info/chapter2_file_management

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management$ ls

demo_folder  project_data

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management$ cd demo_folder

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ pwd

/home/faheemahazin5/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder

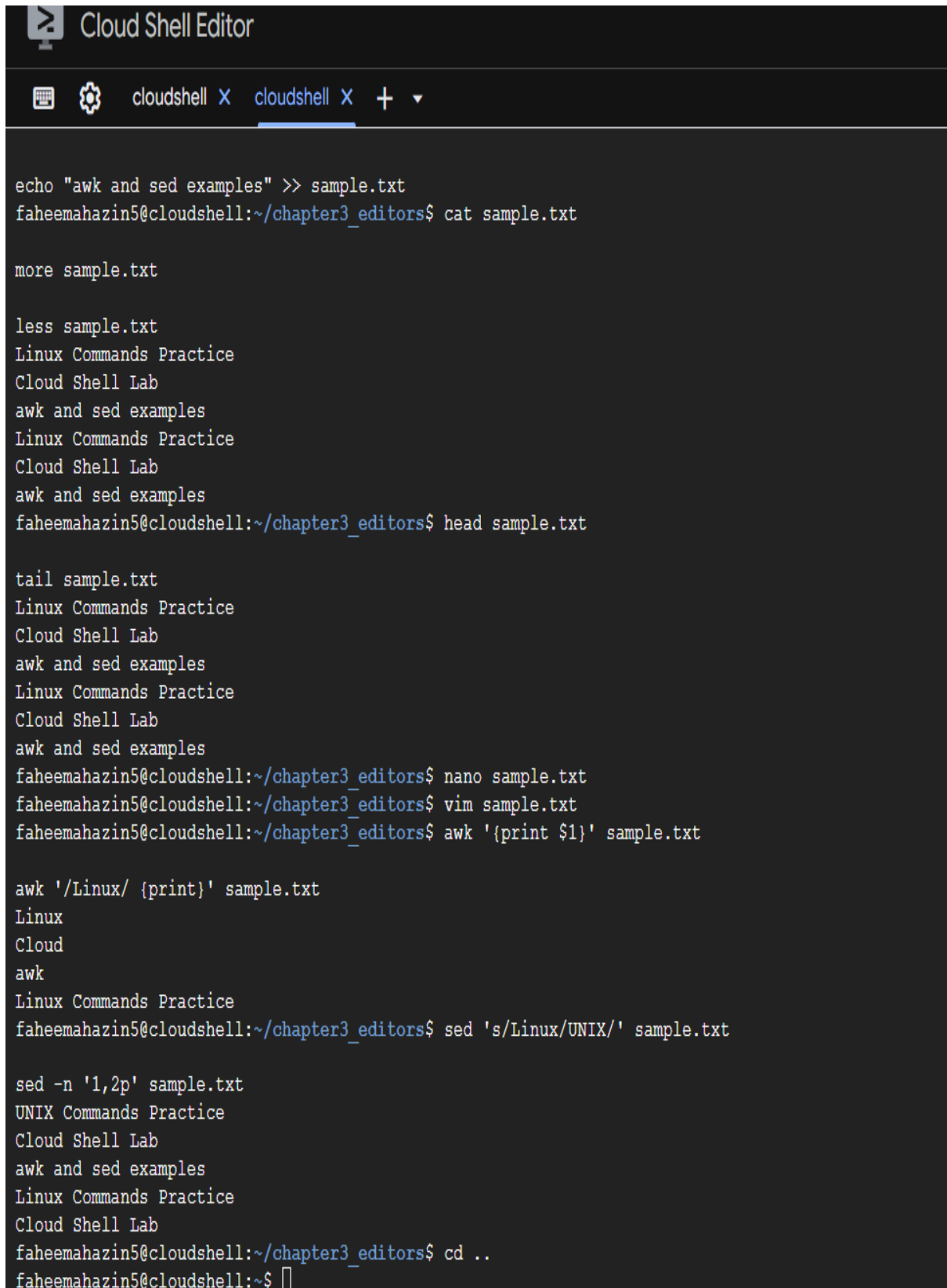
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ touch file1.txt

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ touch file2.txt

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ touch notes.txt

faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ cp file1.txt copy_file1.txt
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ mv file2.txt renamed_file2.txt
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ ln -s notes.txt notes_link
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ rm copy_file1.txt
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ rmdir backup_folder
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info/chapter2_file_management/demo_folder$ cd ..
faheemahazin5@cloudshell:~/linux_lab/chapter1_system_info$
```

CHAPTER 3

A screenshot of a Cloud Shell Editor terminal window. The window has a dark background with a light-colored terminal text. At the top, there is a title bar with a terminal icon and the text "Cloud Shell Editor". Below the title bar is a tab bar with two tabs labeled "cloudshell" and a plus sign icon. The terminal content shows a series of commands and their outputs. The commands include: "echo 'awk and sed examples' >> sample.txt", "cat sample.txt", "more sample.txt", "less sample.txt", "head sample.txt", "tail sample.txt", "nano sample.txt", "vim sample.txt", "awk '{print \$1}' sample.txt", "awk '/Linux/ {print}' sample.txt", "sed 's/Linux/UNIX/' sample.txt", "sed -n '1,2p' sample.txt", and "cd ..". The output of these commands shows the contents of sample.txt, which includes "Linux Commands Practice", "Cloud Shell Lab", and "awk and sed examples". The terminal ends with a prompt "faheemahazin5@cloudshell:~\$".

```
Cloud Shell Editor

cloudshell X cloudshell X + v

echo "awk and sed examples" >> sample.txt
faheemahazin5@cloudshell:~/chapter3_editors$ cat sample.txt

more sample.txt

less sample.txt
Linux Commands Practice
Cloud Shell Lab
awk and sed examples
Linux Commands Practice
Cloud Shell Lab
awk and sed examples
faheemahazin5@cloudshell:~/chapter3_editors$ head sample.txt

tail sample.txt
Linux Commands Practice
Cloud Shell Lab
awk and sed examples
Linux Commands Practice
Cloud Shell Lab
awk and sed examples
faheemahazin5@cloudshell:~/chapter3_editors$ nano sample.txt
faheemahazin5@cloudshell:~/chapter3_editors$ vim sample.txt
faheemahazin5@cloudshell:~/chapter3_editors$ awk '{print $1}' sample.txt

awk '/Linux/ {print}' sample.txt
Linux
Cloud
awk
Linux Commands Practice
faheemahazin5@cloudshell:~/chapter3_editors$ sed 's/Linux/UNIX/' sample.txt

sed -n '1,2p' sample.txt
UNIX Commands Practice
Cloud Shell Lab
awk and sed examples
Linux Commands Practice
Cloud Shell Lab
faheemahazin5@cloudshell:~/chapter3_editors$ cd ..
faheemahazin5@cloudshell:~$
```

CHAPTER 4

```
Cloud Shell Editor

cloudshell x cloudshell x cloudshell x + v

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
faheemahazin5@cloudshell:~$ mkdir chapter4_permissions
cd chapter4_permissions
faheemahazin5@cloudshell:~/chapter4_permissions$ touch secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ ls -l
total 0
-rw-rw-r-- 1 faheemahazin5 faheemahazin5 0 May 14 17:32 secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ chmod 755 secure.txt

chmod 644 secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ umask
0002
faheemahazin5@cloudshell:~/chapter4_permissions$ stat secure.txt
  File: secure.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17      Inode: 131491        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/faheemahazin5)   Gid: ( 1000/faheemahazin5)
Access: 2026-05-14 17:32:11.800203155 +0000
Modify: 2026-05-14 17:32:11.800203155 +0000
Change: 2026-05-14 17:32:37.295640339 +0000
 Birth: 2026-05-14 17:32:11.800203155 +0000
faheemahazin5@cloudshell:~/chapter4_permissions$ sudo apt install acl -y

getfacl secure.txt

setfacl -m u:$USER:rwX secure.txt
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  acl
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 39.4 kB of archives.
After this operation, 197 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 acl amd64 2.3.2-1build1.1 [39.4 kB]
Fetched 39.4 kB in 0s (746 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package acl.
(Reading database ... 121190 files and directories currently installed.)
```

```
Cloud Shell Editor

cloudshell x cloudshell x cloudshell x + v

(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../acl_2.3.2-1build1.1_amd64.deb ...
Unpacking acl (2.3.2-1build1.1) ...
Setting up acl (2.3.2-1build1.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
# file: secure.txt
# owner: faheemahazin5
# group: faheemahazin5
user::rw-
group::r--
other::r--

faheemahazin5@cloudshell:~/chapter4_permissions$ sudo apt install acl -y

getfacl secure.txt

setfacl -m u:$USER:rwX secure.txt
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
acl is already the newest version (2.3.2-1build1.1).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
# file: secure.txt
# owner: faheemahazin5
# group: faheemahazin5
user::rw-
user:faheemahazin5:rwX
group::r--
mask::rwX
other::r--

faheemahazin5@cloudshell:~/chapter4_permissions$ lsattr secure.txt

sudo chatter +i secure.txt

lsattr secure.txt
-----e----- secure.txt
---i-----e----- secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ sudo chatter -i secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ cd ..
faheemahazin5@cloudshell:~$
```

CHAPTER 5

```
Cloud Shell Editor

cloudshell x cloudshell x cloudshell x + v

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
faheemahazin5@cloudshell:~$ mkdir chapter4_permissions
cd chapter4_permissions
faheemahazin5@cloudshell:~/chapter4_permissions$ touch secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ ls -l
total 0
-rw-rw-r-- 1 faheemahazin5 faheemahazin5 0 May 14 17:32 secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ chmod 755 secure.txt

chmod 644 secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ umask
0002
faheemahazin5@cloudshell:~/chapter4_permissions$ stat secure.txt
  File: secure.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17      Inode: 131491       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/faheemahazin5)   Gid: ( 1000/faheemahazin5)
Access: 2026-05-14 17:32:11.800203155 +0000
Modify: 2026-05-14 17:32:11.800203155 +0000
Change: 2026-05-14 17:32:37.295640339 +0000
 Birth: 2026-05-14 17:32:11.800203155 +0000
faheemahazin5@cloudshell:~/chapter4_permissions$ sudo apt install acl -y

getfacl secure.txt

setfacl -m u:$USER:rwX secure.txt
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  acl
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 39.4 kB of archives.
After this operation, 197 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 acl amd64 2.3.2-1build1.1 [39.4 kB]
Fetched 39.4 kB in 0s (746 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package acl.
(Reading database ... 121190 files and directories currently installed.)
```

```
Cloud Shell Editor

cloudshell x cloudshell x cloudshell x + v

(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../acl_2.3.2-1build1.1_amd64.deb ...
Unpacking acl (2.3.2-1build1.1) ...
Setting up acl (2.3.2-1build1.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
# file: secure.txt
# owner: faheemahazin5
# group: faheemahazin5
user::rw-
group::r--
other::r--

faheemahazin5@cloudshell:~/chapter4_permissions$ sudo apt install acl -y

getfacl secure.txt

setfacl -m u:$USER:rwX secure.txt
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
acl is already the newest version (2.3.2-1build1.1).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
# file: secure.txt
# owner: faheemahazin5
# group: faheemahazin5
user::rw-
user:faheemahazin5:rwX
group::r--
mask::rwX
other::r--

faheemahazin5@cloudshell:~/chapter4_permissions$ lsattr secure.txt

sudo chattr +i secure.txt

lsattr secure.txt
-----e----- secure.txt
----i-----e----- secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ sudo chattr -i secure.txt
faheemahazin5@cloudshell:~/chapter4_permissions$ cd ..
faheemahazin5@cloudshell:~$
```

```

To set your Cloud Platform project in this session use 'gcloud config set project [PROJECT_ID]'.
You can view your projects by running 'gcloud projects list'.
faheemahazin5@cloudshell:~$ mkdir chapter5_networking
cd chapter5_networking
faheemahazin5@cloudshell:~/chapter5_networking$ sudo apt install net-tools traceroute dnsutils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
net-tools is already the newest version (2.10-0.1ubuntu4.4).
dnsutils is already the newest version (1:9.18.39-0ubuntu0.24.04.3).
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 0s (2,228 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package traceroute.
(Reading database ... 121200 files and directories currently installed.)
Preparing to unpack .../traceroute_1:2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute.1.gz because associated file /usr/share/man/man1/traceroute.db.1.gz (of link group traceroute) doesn't exist
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute6.1.gz because associated file /usr/share/man/man1/traceroute6.db.1.gz (of link group traceroute6) doesn't exist
update-alternatives: using /usr/bin/ift.db to provide /usr/bin/ift (ift) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/ift.1.gz because associated file /usr/share/man/man1/ift.db.1.gz (of link group ift) doesn't exist
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceproto.1.gz because associated file /usr/share/man/man1/traceproto.db.1.gz (of link group traceproto) doesn't exist
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man3/tcptraceroute.8.gz because associated file /usr/share/man/man3/tcptraceroute.db.8.gz (of link group tcptraceroute) doesn't exist
Processing triggers for man-db (2.12.0-4build2) ...
faheemahazin5@cloudshell:~/chapter5_networking$ ifconfig > network_info.txt

ip a >> network_info.txt

ping -c 4 google.com

```

```

netstat -tulnp

ss -tulnp

traceroute google.com

wget https://example.com

curl https://example.com

nslookup google.com

dig google.com
PING google.com (142.251.10.113) 56(84) bytes of data.
 64 bytes from sd-in-f113.1e100.net (142.251.10.113): icmp_seq=1 ttl=114 time=0.953 ms
 64 bytes from sd-in-f113.1e100.net (142.251.10.113): icmp_seq=2 ttl=114 time=0.539 ms
 64 bytes from sd-in-f113.1e100.net (142.251.10.113): icmp_seq=3 ttl=114 time=0.561 ms
 64 bytes from sd-in-f113.1e100.net (142.251.10.113): icmp_seq=4 ttl=114 time=0.532 ms

--- google.com ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3030ms
 rtt min/avg/max/mdev = 0.532/0.646/0.953/0.177 ms
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)

```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State	PID/Program name
tcp	0	0	127.0.0.1:33107	0.0.0.0:*	LISTEN	-
tcp	0	0	127.0.0.1:8998	0.0.0.0:*	LISTEN	-
tcp	0	0	127.0.0.1:43331	0.0.0.0:*	LISTEN	-
tcp	0	0	127.0.0.1:38723	0.0.0.0:*	LISTEN	-
tcp	0	0	0.0.0.0:971	0.0.0.0:*	LISTEN	-
tcp	0	0	0.0.0.0:22	0.0.0.0:*	LISTEN	-
tcp6	0	0	:::922	:::*	LISTEN	-
tcp6	0	0	:::900	:::*	LISTEN	-
tcp6	0	0	:::981	:::*	LISTEN	-
tcp6	0	0	:::980	:::*	LISTEN	-
tcp6	0	0	:::970	:::*	LISTEN	-
tcp6	0	0	:::22	:::*	LISTEN	-

```

Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
tcp    LISTEN  0       128    127.0.0.1:33107    0.0.0.0:*
tcp    LISTEN  0       5      127.0.0.1:8998     0.0.0.0:*

```

```

Netid  State  Recv-Q  Send-Q    Local Address:Port    Peer Address:Port    Process
tcp    LISTEN  0        128      127.0.0.1:33107      0.0.0.0:*
tcp    LISTEN  0         5       127.0.0.1:8998      0.0.0.0:*
tcp    LISTEN  0        511     127.0.0.1:43331     0.0.0.0:*
tcp    LISTEN  0        511     127.0.0.1:38723     0.0.0.0:*
tcp    LISTEN  0        511      0.0.0.0:971        0.0.0.0:*
tcp    LISTEN  0        128      0.0.0.0:22         0.0.0.0:*
tcp    LISTEN  0       4096      *:922              *:
tcp    LISTEN  0       4096      *:900              *:
tcp    LISTEN  0       4096      *:981              *:
tcp    LISTEN  0       4096      *:980              *:
tcp    LISTEN  0       4096      *:970              *:
tcp    LISTEN  0        128      [::]:22            [::]:*

traceroute to google.com (142.251.10.113), 30 hops max, 60 byte packets
 1  10.88.0.1 (10.88.0.1)  0.033 ms  0.009 ms  0.007 ms
 2  * 209.85.250.37 (209.85.250.37)  2.301 ms *
 3  * * *
 4  * * *
 5  * * *
 6  * * 142.251.52.243 (142.251.52.243)  0.445 ms
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * sd-in-fl113.1e100.net (142.251.10.113)  0.606 ms
--2026-05-14 17:42:12-- https://example.com/
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:10::ac42:93f3, ...
Connecting to example.com (example.com)|172.66.147.243|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html          [ <=>          ]    528  --.-KB/s   in 0s

2026-05-14 17:42:12 (78.9 MB/s) - 'index.html' saved [528]

```

```

Address: 142.251.10.101
Name: google.com
Address: 142.251.10.138
Name: google.com
Address: 142.251.10.139
Name: google.com
Address: 2404:6800:4003:c11::66
Name: google.com
Address: 2404:6800:4003:c11::8b
Name: google.com
Address: 2404:6800:4003:c11::8a
Name: google.com
Address: 2404:6800:4003:c11::65

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17577
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                287     IN      A      142.251.10.113
google.com.                287     IN      A      142.251.10.100
google.com.                287     IN      A      142.251.10.102
google.com.                287     IN      A      142.251.10.101
google.com.                287     IN      A      142.251.10.138
google.com.                287     IN      A      142.251.10.139

;; Query time: 1 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 17:42:12 UTC 2026
;; MSG SIZE rcvd: 135

faheemahazin5@cloudshell:~/chapter5_networking$ cd ..
faheemahazin5@cloudshell:~$

```

CHAPTER 6

To set your Cloud Platform project in this session use `gcloud config set project [PROJECT\_ID]`.

You can view your projects by running `gcloud projects list`.

```
faheemahazin5@cloudshell:~$ mkdir chapter6_process
```

```
cd chapter6_process
```

```
faheemahazin5@cloudshell:~/chapter6_process$ ps
```

```
ps aux
```

	PID	TTY		TIME	CMD
	3691	pts/11		00:00:00	bash
	4067	pts/11		00:00:00	ps

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.0	4324	3248	?	Ss	16:21	0:00	/bin/bash /google/scripts/o
syslog	9	0.0	0.0	222140	3908	?	Ssl	16:21	0:00	/usr/sbin/rsyslogd
root	25	0.0	0.1	38768	27928	?	S	16:21	0:01	/usr/bin/python /usr/bin/su
root	26	0.0	0.0	6196	2716	?	S	16:21	0:00	logger -t supervisord
root	67	0.0	0.0	12024	4152	?	S<s	16:21	0:00	sshd: /usr/sbin/sshd -p 22
root	347	0.0	0.5	2096096	86148	?	Sl	16:21	0:01	/usr/bin/dockerd -p /var/ru
root	440	0.0	0.0	1234524	12548	?	Sl	16:21	0:01	/google/devshell/editor/pro
root	441	0.0	0.0	16328	6372	?	S	16:21	0:00	sudo /google/devshell/tmux-
root	446	0.0	0.0	1226100	2752	?	Sl	16:21	0:00	/google/devshell/tmux-agent
root	467	0.1	0.2	1867212	46528	?	Ssl	16:21	0:07	containerd --config /var/ru
root	682	0.0	0.0	2696	1564	?	S	16:21	0:00	sleep infinity
root	735	0.0	0.0	9196	3932	?	S	16:23	0:00	runuser -l faheemahazin5 -c
faheema+	736	0.0	0.0	4324	3380	?	Ss	16:23	0:00	sh ./bin/codeoss-cloudshell
faheema+	757	0.1	0.9	11868972	151400	?	Sl	16:23	0:07	/google/devshell/editor/cod
root	1202	0.0	0.0	13996	9708	?	S<s	16:37	0:00	sshd: faheemahazin5 [priv]
faheema+	1206	0.0	0.0	14256	6556	?	S<	16:37	0:00	sshd: faheemahazin5@pts/0
faheema+	1207	0.0	0.0	7340	3784	pts/0	S<s	16:37	0:00	bash -c (sudo touch /var/ru
faheema+	1210	0.0	0.0	7604	4480	pts/0	S<+	16:37	0:00	/bin/bash --norc --noprofil
faheema+	1218	0.0	0.0	7948	3568	?	S<s	16:37	0:00	tmux new-session -A -D -n c
faheema+	1219	0.0	0.0	8928	5964	pts/2	S<s+	16:37	0:00	-bash
faheema+	1588	1.0	6.5	44821276	1080016	?	Sl	16:37	0:47	/google/devshell/editor/cod
faheema+	1599	0.0	0.3	1048516	62500	?	Sl	16:37	0:00	/google/devshell/editor/cod
faheema+	1619	0.0	0.3	1016392	62140	?	Sl	16:37	0:00	/google/devshell/editor/cod
faheema+	1636	0.1	1.3	32890804	216920	?	Sl	16:38	0:07	/google/devshell/editor/cod
faheema+	1643	0.1	3.7	1853604	616556	?	Sl	16:38	0:08	/google/devshell/editor/cod
root	1719	0.0	0.0	13996	9812	?	S<s	17:02	0:00	sshd: faheemahazin5 [priv]
faheema+	1721	0.0	0.0	14256	6384	?	S<	17:03	0:00	sshd: faheemahazin5@pts/1
faheema+	1722	0.0	0.0	7340	3604	pts/1	S<s+	17:03	0:00	bash -c if [-f /google/dev
faheema+	1723	0.0	0.0	7340	3736	pts/1	S<+	17:03	0:00	/bin/bash /google/devshell/
faheema+	1724	0.0	0.0	6880	3400	pts/1	S<+	17:03	0:00	tmux new-session -A -D -n c

```

faheemahazin5@cloudshell:~/chapter6_process$ pgrep bash
1
1207
1210
1219
1722
1725
2241
2244
2698
2701
3195
3198
3688
3691
faheemahazin5@cloudshell:~/chapter6_process$ sleep 300 &
[1] 4078
faheemahazin5@cloudshell:~/chapter6_process$ jobs
[1]+  Running                  sleep 300 &
faheemahazin5@cloudshell:~/chapter6_process$ fg
sleep 300
^Z
[1]+  Stopped                  sleep 300
faheemahazin5@cloudshell:~/chapter6_process$ bg
[1]+ sleep 300 &
faheemahazin5@cloudshell:~/chapter6_process$ pkill sleep

killall sleep
pkill: killing pid 682 failed: Operation not permitted
[1]+  Terminated              sleep 300
sleep(682): Operation not permitted
sleep: no process found
faheemahazin5@cloudshell:~/chapter6_process$ nice -n 10 sleep 200 &
[1] 4097
faheemahazin5@cloudshell:~/chapter6_process$ ps aux | grep sleep
root      1  0.0  0.0  4324 3248 ?        Ss   16:21   0:00 /bin/bash /google/scripts/onrun.sh sleep infinity
root      682  0.0  0.0  2696 1564 ?        S    16:21   0:00 sleep infinity
faheema+  4097  0.0  0.0  5688 2236 pts/11   S    17:54   0:00 sleep 200
faheema+  4101  0.0  0.0  6548 2420 pts/11  S<+  17:54   0:00 grep --color=auto sleep

```

```

faheemahazin5@cloudshell:~/chapter6_process$ pgrep sleep
682
4513
faheemahazin5@cloudshell:~/chapter6_process$ renice 5 4513
4513 (process ID) old priority -10, new priority 5
faheemahazin5@cloudshell:~/chapter6_process$ █

```

CHAPTER 7

```
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
faheemahazin5@cloudshell:~$ mkdir chapter7_disk_management
cd chapter7_disk_management
faheemahazin5@cloudshell:~/chapter7 disk management$ df -h
```

```
du -sh .
Filesystem      Size  Used Avail Use% Mounted on
overlay          95G   73G   22G   78% /
tmpfs           64M    0   64M    0% /dev
/dev/sdal        95G   73G   22G   78% /root
/dev/disk/by-id/google-home-part1 4.8G  152M   4.4G   4% /home
/dev/root        2.0G   1.3G   715M  64% /usr/lib/modules
shm             64M    0   64M    0% /dev/shm
4.0K
faheemahazin5@cloudshell:~/chapter7 disk management$ lsblk
```

```
blkid
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda         8:0    0 100G  0 disk
├─sda1      8:1    0 95.8G  0 part /run/google/devshell
│           /var/lib/docker
│           /etc/ssh/keys
│           /var/config/tmux
│           /etc/resolv.conf
│           /etc/hostname
│           /etc/hosts
│           /root
├─sda2      8:2    0  16M  0 part
├─sda3      8:3    0   2G  0 part
├─sda4      8:4    0  16M  0 part
├─sda5      8:5    0   2G  0 part
├─sda6      8:6    0 512B  0 part
├─sda7      8:7    0 512B  0 part
├─sda8      8:8    0  16M  0 part
├─sda9      8:9    0 512B  0 part
├─sda10     8:10   0 512B  0 part
├─sda11     8:11   0   8M  0 part
└─sda12     8:12   0  32M  0 part
sdb         8:16    0   5G  0 disk
└─sdb1      8:17   0   5G  0 part /home
```

```
faheemahazin5@cloudshell:~/chapter7 disk management$ sudo fdisk -l
```

```

sudo parted -l
sudo: fdisk: command not found
sudo: parted: command not found
faheemahazin5@cloudshell:~/chapter7_disk_management$ sudo tune2fs -l /dev/sda1
tune2fs 1.47.0 (5-Feb-2023)
Filesystem volume name:   STATE
Last mounted on:          /var
Filesystem UUID:          9f2dbcb8-9534-4465-bea2-491409599ab8
Filesystem magic number:  0xEF53
Filesystem revision #:    1 (dynamic)
Filesystem features:      has_journal ext_attr resize_inode dir_index filetype needs_recovery extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags:         signed_directory_hash
Default mount options:    user_xattr acl
Filesystem state:         clean
Errors behavior:          Remount read-only
Filesystem OS type:       Linux
Inode count:              6258720
Block count:              25126395
Reserved block count:     0
Overhead clusters:        418418
Free blocks:              5630074
Free inodes:              4791093
First block:              0
Block size:               4096
Fragment size:            4096
Group descriptor size:    64
Reserved GDT blocks:      609
Blocks per group:         32768
Fragments per group:      32768
Inodes per group:         8160
Inode blocks per group:   510
Flex block group size:    16
Filesystem created:       Fri May  8 10:15:28 2026
Last mount time:          Thu May 14 09:10:20 2026
Last write time:          Thu May 14 17:39:17 2026
Mount count:              6
Maximum mount count:      -1
Last checked:             Thu Nov 19 11:00:00 2009
Check interval:           0 (<none>)

```

CHAPTER 8

```
LI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
faheemahazin5@cloudshell:~$ mkdir chapter8_performance
cd chapter8_performance
faheemahazin5@cloudshell:~/chapter8_performance$ sudo apt install sysstat dstat glances -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'pcp' instead of 'dstat'
The following additional packages will be installed:
  blt fonts-lyx gawk isympy-common isympy3 javascript-common libblas3 libdouble-conversion3
  libimagequant0 libjs-jquery-ui liblapack3 liblbfgsb0 libopenjp2-7 libpcp-archive1t64
  libpcp-gui2t64 libpcp-import1t64 libpcp-mmvt64 libpcp-pmda-perl libpcp-pmda3t64
  libpcp-trace2t64 libpcp-web1t64 libpcp3t64 libpfm4 libqhull-r8.0 libraqm0 libsigsegv2 libtcl8.6
  libtk8.6 libwebpdemux2 libwebpmux3 libxft2 libxss1 lm-sensors pcp-conf python-matplotlib-data
  python3-appdirs python3-attr python3-bottle python3-brotli python3-bs4 python3-certifi
  python3-chardet python3-contourpy python3-cssselect python3-cycler python3-dateutil
  python3-decorator python3-defusedxml python3-docker python3-fonttools python3-fs
  python3-html5lib python3-idna python3-influxdb python3-kiwisolver python3-lxml python3-lz4
  python3-matplotlib python3-mpmath python3-msgpack python3-netifaces python3-numpy
  python3-olefile python3-pcp python3-pil python3-pil.imagetk python3-ply python3-psutil
  python3-pyasnl python3-pycryptodome python3-pysmi python3-pysnmp4 python3-pystache
  python3-requests python3-scipy python3-soupsieve python3-sympy python3-tk python3-tz
  python3-ufolib2 python3-urjson python3-unicodedata2 python3-urllib3 python3-webencodings
  python3-websocket tk8.6-blt2.5 unicode-data
Suggested packages:
  blt-demo gawk-doc glances-doc python3-pynvml apache2 | lighttpd | httpd libjs-jquery-ui-docs
  tcl8.6 tk8.6 fancontrol read-edid i2c-tools pcp-gui libpcp-import-perl bpftool python3-bpfcc
  python-attr-doc python-cycler-doc python3-genshi python-lxml-doc cm-super-minimal dvipng ffmpeg
  fonts-staypuft ghostscript glib2.0-gtk-3.0 inkscape ipython3 librsync2-common python3-cairocffi
  python3-gi-cairo python3-gobject python3-pyqt5 python3-sip python3-tornado texlive-extra-utils
  texlive-latex-extra python3-mpmath-doc python3-gmpy2 gfortran python3-pytest python-pil-doc
  python-ply-doc python3-openssl python3-socks python-requests-doc python-scipy-doc
  texlive-fonts-extra python-sympy-doc tix python3-tk-dbg isag
The following NEW packages will be installed:
  blt fonts-lyx gawk glances isympy-common isympy3 javascript-common libblas3
  libdouble-conversion3 libimagequant0 libjs-jquery-ui liblapack3 liblbfgsb0 libopenjp2-7
  libpcp-archive1t64 libpcp-gui2t64 libpcp-import1t64 libpcp-mmvt64 libpcp-pmda-perl
  libpcp-pmda3t64 libpcp-trace2t64 libpcp-web1t64 libpcp3t64 libpfm4 libqhull-r8.0 libraqm0
  libsigsegv2 libtcl8.6 libtk8.6 libwebpdemux2 libwebpmux3 libxft2 libxss1 lm-sensors pcp
```

```
faheemahazin5@cloudshell:~/chapter8_performance$ iostat

vmstat

sar

mpstat

dstat

glances
Linux 6.6.137+ (cs-782581850354-default)      05/14/2026      _x86_64_      (4 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.45    0.00    0.29    0.03    0.02   99.21

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
dm-0               0.24        12.99         0.00         0.00      435300         0         0
sda               2.72        51.50        31.56         0.00     1725415     1057281         0
sdb               0.04         3.95         0.16         0.00      132298       5276         0

procs -----memory----- ---swap-- ----io-----system-- -----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st gu
 1 0   0 11447384 123296 2416352 0 0 55 31 782 3 0 0 99 0 0 0
Cannot open /var/log/sysstat/sa14: No such file or directory
Please check if data collecting is enabled
Linux 6.6.137+ (cs-782581850354-default)      05/14/2026      _x86_64_      (4 CPU)

06:28:41 PM CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
06:28:41 PM all    0.45    0.00    0.27    0.03    0.00    0.02    0.02    0.00    0.00   99.21
You did not select any stats, using -cdngy by default.
Error Parsing ASCII PMNS: Cannot open "/var/lib/pcp/pmns/root"
Failed to register derived metric:
PMNS not accessible <pcp.pmap.c_uint_Array_1 object at 0x7a85665b4dd0>.

faheemahazin5@cloudshell:~/chapter8_performance$
```

CHAPTER 9

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
faheemahazin5@cloudshell:~$ mkdir chapter9_user_management
cd chapter9_user_management
mkdir: cannot create directory 'chapter9_user_management': File exists
faheemahazin5@cloudshell:~/chapter9_user_management$ sudo useradd testuser
faheemahazin5@cloudshell:~/chapter9_user_management$ sudo usermod -aG sudo testuser
faheemahazin5@cloudshell:~/chapter9_user_management$ groups

groups testuser
faheemahazin5 adm sudo docker
testuser : testuser sudo
faheemahazin5@cloudshell:~/chapter9_user_management$ sudo groupadd developers
faheemahazin5@cloudshell:~/chapter9_user_management$ sudo usermod -aG developers testuser
faheemahazin5@cloudshell:~/chapter9_user_management$ su testuser
```

```
faheemahazin5@cloudshell:~/chapter8_performance$ iostat

vmstat

sar

mpstat

dstat

glances
Linux 6.6.137+ (cs-782581850354-default)      05/14/2026      _x86_64_      (4 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.45    0.00   0.29   0.03    0.02   99.21

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
dm-0               0.24       12.99         0.00         0.00     435300         0         0
sda               2.72       51.50        31.56         0.00    1725415    1057281         0
sdb               0.04        3.95         0.16         0.00     132298      5276         0

procs -----memory----- ---swap-- ----io---- -system-- -----cpu-----
 r b  swpd   free   buff  cache   si   so    bi   bo    in   cs us sy id wa st gu
  1 0      0 11447384 123296 2416352    0    0    55   31  782    3  0  0 99  0  0  0
Cannot open /var/log/sysstat/sa14: No such file or directory
Please check if data collecting is enabled
Linux 6.6.137+ (cs-782581850354-default)      05/14/2026      _x86_64_      (4 CPU)

06:28:41 PM CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal   %guest   %gnice   %idle
06:28:41 PM all      0.45    0.00    0.27   0.03    0.00    0.02   0.02   0.00   0.00   99.21
You did not select any stats, using -cdngy by default.
Error Parsing ASCII PMNS: Cannot open "/var/lib/pcp/pmns/root"
Failed to register derived metric:
PMNS not accessible <pcp.pmapi.c_uint_Array_1 object at 0x7a85665b4dd0>.

faheemahazin5@cloudshell:~/chapter8_performance$ sudo userdel testuser
faheemahazin5@cloudshell:~/chapter8_performance$ sudo groupdel developers
faheemahazin5@cloudshell:~/chapter8_performance$
```

CHAPTER 10

LI.

To set your Cloud Platform project in this session use ``gcloud config set project [PROJECT_ID]``.

You can view your projects by running ``gcloud projects list``.

```
faheemahazin5@cloudshell:~$ mkdir chapter10_compression
```

```
cd chapter10_compression
```

```
faheemahazin5@cloudshell:~/chapter10_compression$ touch data.txt
```

```
echo "Linux Compression Practice" > data.txt
```

```
faheemahazin5@cloudshell:~/chapter10_compression$ bzip2 data.txt
```

```
bunzip2 data.txt.bz2
```

```
faheemahazin5@cloudshell:~/chapter10_compression$ xz data.txt
```

```
unxz data.txt.xz
```

```
faheemahazin5@cloudshell:~/chapter10_compression$ tar -cvf archive.tar data.txt
```

```
tar -xvf archive.tar
```

```
data.txt
```

```
data.txt
```

```
faheemahazin5@cloudshell:~/chapter10_compression$ zip archive.zip data.txt
```

```
unzip archive.zip
```

```
adding: data.txt (stored 0%)
```

```
Archive: archive.zip
```

```
replace data.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: lzma data.txt
```